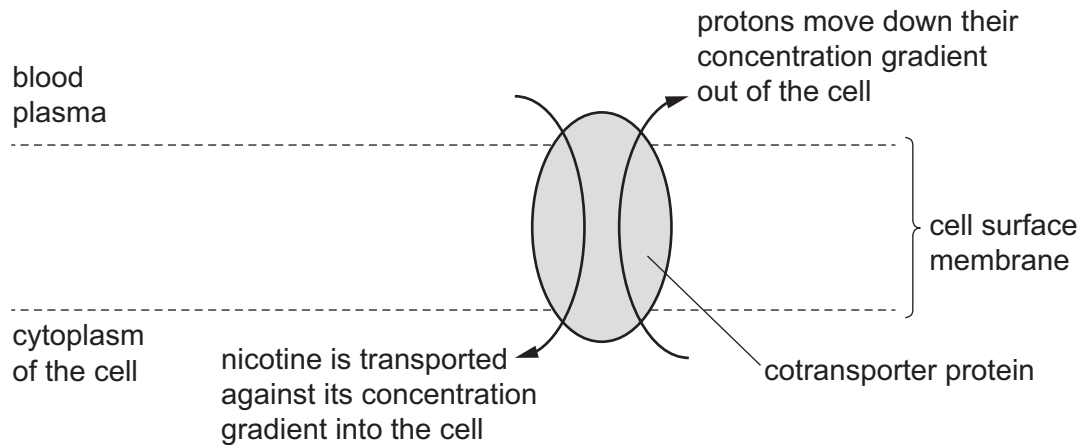


- 13** The diagram shows how nicotine is transported from the blood plasma into a cell using a type of cotransporter mechanism.



In the phloem tissue, there is a cotransporter mechanism that moves sucrose into the cytoplasm of a companion cell.

Which statement correctly describes a similarity between the cotransport of nicotine and the cotransport of sucrose?

- A** The cotransporter proteins generate a proton gradient by moving protons out of the cell by active transport.
- B** The protons are transported through the cotransporter proteins by facilitated diffusion.
- C** The protons move through the cotransporter proteins in the opposite direction to the movement of nicotine and sucrose.
- D** The cotransporter proteins use energy from ATP to transport protons with nicotine and sucrose.